MILITARY SPECIFICATION

MATS, LANDING, ALUMINUM: HEAVY DUTY, AND ANCILLARY ITEMS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a heavy-duty expedient airfield surfacing system consisting of landing mats and ancillary items.

1.2 Classification. The components, mats, and ancillary items, shall be of the following types as specified (see 6.2):

Landing mats

Type I	- Full panel.
Type II	- Half panel.
Type III	- Repair panel.

Ancillary items

Ancillary 1	- Adapter, access.
Ancillary 2	- Anchor attachment.
Ancillary 3	- Edge anchor assembly.
Ancillary 4	- Bar, locking.
Ancillary 5	- Starting connector.
Ancillary 6	- Turn adapter, female.
Ancillary 7	- Turn adapter, male.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

FF-N-105	- Nails, Brads, Staples, and Spikes: Wire, Cut and Wrought.

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NN-P-530	- Plywood, Flat Panel.
QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.

Military

MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-A-8625	- Anodic Coatings, For Aluminum and Aluminum Alloys.
MIL-C-81346	- Compound, Deck Covering, Nonslip, Lightweight for Aluminum Alloy Landing Mats.

STANDARDS

Federal

FED. STD. No. 595	- Colors.

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-621	- Test Method for Pavement Subgrade, Subbase and Base Course Materials.
MIL-STD-731	- Quality of Wood Members for Containers and Pallets.

DRAWINGS

ME

TA13220E5650	- Landing Mats Assembly.
C13216E9236	- Edge Anchor Assembly.
D13220E5630	- Mat, Landing, Full Mat, Heavy Duty.
D13220E5631	- Mat, Landing, Half Mat, Heavy Duty.

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C13220E5633	- Bar, Locking.
C13220E5634	- Connector, Starting.
D13220E5635	- Adapter, Access.
D13220E5636	- Turn Adapter, Male/I-Lock.
D13220E5637	- Turn Adapter, Female/I-Lock.
C13220E5639	- Anchor Attachment, I-Lock.
D13220E5640	- Repair Panel Assembly Heavy Duty.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

THE ALUMINUM ASSOCIATION

Aluminum Standards and Data.

(Application for copies should be addressed to the Aluminum Association, 420 Lexington, Avenue, New York, NY 10017.)

3. REQUIREMENTS

3.1 Description. The landing mat and ancillary items shall be in accordance with Top Assembly TA13220E5650, and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the supplier shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the landing mat system. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

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3.2 First article (preproduction model). The supplier shall furnish one or more landing mats or ancillary items as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods will produce landing mats or ancillary items that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the supplier subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the supplier shall furnish to the Government one or more of each type of mat and ancillary item for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.5 Interchangeability. When tested as specified in 4.6.2.1, the landing mat shall assemble and be fully interchangeable with like mats.

3.6 Structural capability and reliability. When tested as specified in 4.6.2.2 the landing mat, when placed on a subgrage with a CBR of 4, shall have a 0.90 probability of withstanding 1,000 coverages of a 50,000 pound wheel load that has a tire pressure of 250 psi and a tire footprint of 200 square inches.

3.7 Identification marking. The mats shall be identified in accordance with MIL-STD-130, and as specified herein.

3.8 Finishing.

3.8.1 Chemical conversion coating. The mat shall be cleaned and given a chemical conversion coating in accordance with MIL-C-5541, Class 1A.

3.8.2 Nonskid wearing surface. After the mat has been given a chemical conversion coating, the upper surface of the mat shall be coated with nonskid coating conforming to MIL-C-81346, Grade 1 or 2 as applicable, except that the color shall be Color No. 34087 in accordance with FED. STD. No. 595. The thickness of the nonskid wearing surface shall be not less than 5 and not more than 10 mils thick, and cover the entire wearing surface. Nonskid compound shall not be applied to the connector rails.

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3.8.3 Lower surface of mat. The lower surface of the mat shall be coated as specified in 3.8.1, and no painting is required.

3.8.4 Ancillary items. Ancillary items, except for locking bars, shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A. Locking bars shall be anodized in accordance with MIL-A-8625, Type I or II, Class 1.

3.9 Metal stamping. Each mat and ancillary item, except for locking bars, shall be metal stamped on an exposed position near one end of component with the National Stock Number shown on the drawing. Numerals shall be 1/2 inch where space permits. On small components, metal stamping shall be as large as practicable.

3.10 Type I landing mat. Type I landing mat shall be as shown on Drawing D13220E5630, and as specified herein. The mat shall weigh not more than 116 pounds plus or minus 3 pounds.

3.11 Type II landing mat. Type II landing mat shall be as shown on Drawing D13220E5631, and as specified herein. The mat shall weigh not more than 60 pounds plus or minus 2 pounds.

3.12 Type III landing mat. Type III landing mat shall be as shown on Drawing D13220E5640 and as specified herein. The mat shall weigh not more than 141 pounds plus or minus 2 pounds.

3.13 Ancillary items. Ancillary items shall be as shown on the following drawings, and as specified herein:

Item	Drawing	Nomenclature
Ancillary 1	-D13220E5635	Adapter, Access.
Ancillary 2	-C13220E5639	Anchor Attachment.
Ancillary 3	-C13216E9236	Edge Anchor Assembly
Ancillary 4	-C13220E5633	Bar, Locking.
Ancillary 5	-C13220E5634	Starting connector.
Ancillary 6	-D13220E5637	Turn adapter, female.
Ancillary 7	-D13220E5636	Turn adapter, male.

3.14 Workmanship.

3.14.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods

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that will not cause injury to the metal. Corners shall be square and true. All bends shall be made with controlled means to insure uniformity of size and shape. Aluminum alloys shall be handled and stored in accordance with the procedure outlined in the aluminum Associations's Handbook, Aluminum Standards and Data. Alloys specified on the drawings shall not be permitted to come into contact with corrosive materials during storage. Cutting of aluminum alloys shall be accomplished by sawing, routing, or machining methods. Flame- or arc-cutting, or shearing, shall not be permitted.

3.14.2 Welding procedure. Prior to the production fabrication of any weldment, the supplier shall prepare and submit to the contracting officer for approval an isometric or perspective drawing showing the location of each joint and the joint welding procedures to cover all welding, including a general outline for the repair of base metal and welded joints. The procedures shall be qualified in accordance with ASME Boiler and Pressure Vessel Code, Section IX. All welds that are to be qualified shall be made on the same alloy, gage, and temper of material, using the filler alloy specified on the drawings and the same procedure and positions as will be used in production. Weld procedures which have been qualified under other equivalent recognized codes may be considered qualified for the purposes of this specification, provided they are approved by the contracting officer. Any change in procedure during fabrication shall require requalification in accordance with the requirements of the ASME Boiler and Pressure Vessel Code, Section IX.

3.14.3 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, oxide film, and other foreign matter. Welds shall be free from defects such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, porosity, lack of penetration, lack of prescribed fit, slag inclusion, or spatter.

3.14.4 Welders. Before assigning any welder to welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by the ASME Boiler and Pressure Vessel Code, Section IX, for the type of welding operations to be performed. Test specimens shall be made on the same alloy, gage, and temper of material with the same filler alloy and in the same position as will be used in production. Welders who have been similarly qualified under other recognized codes may be considered qualified for the purposes of this specification, provided they are approved by the contracting officer. Operators of automatic welding equipment shall be considered welders.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.5 and 6.6).

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.8).
(e) Inspection of preparation for delivery (see 4.9).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction mats and ancillary items, as applicable, shall be examined for the defects specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction mats and ancillary items, as applicable, shall be tested as specified in 4.6.2. Failure of any test shall be cause for rejection.

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4.4 Initial production inspection. When specified (see 3.3), initial production mats and ancillary items, as applicable, in the quantities required, will be selected at random by the Government from the first mats or ancillary items being produced from production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of initial production mats or ancillary items shall not exclude the remaining mats or ancillary items from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of initial production mats or ancillary items to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production mats or ancillary items and shall be cause for refusal by the Government to continue acceptance of production mats, ancillary items, or components until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on mats or ancillary items previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all mats or ancillary items accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Samples. Samples selected in accordance with 4.5.1 shall be examined for the defects specified in 4.6.1. AQL shall be 2.5 percent defective.

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4.5.3 Tests. The landing mats and listed ancillary items shall be tested as specified in 4.6.2.

4.6 Inspection procedure.

4.6.1 Examination. The mats and ancillary items shall be examined as specified herein for the following defects:

Major

101. Any part or component not in accordance with the SQAP requirements as shown on the drawings.
102. Materials not as specified.
103. Identification marking not as specified.
104. Finishing not as specified.
105. Nonskid compound not applied to the mat wearing surface as specified.
106. Connector rails not free from nonskid compound as specified.
107. Chemical conversion not applied to bottom of mat as specified.
108. Ancillary items not anodized as specified.
109. Metal stamping not as specified.
110. Weight of mats not as specified.
111. Metal fabrication not as specified.
112. Nonconformance to approved welding procedure.
113. Welds not free from the specified defects.
114. Welders not certified as specified.
115. Workmanship not as specified.

4.6.2 Tests.

4.6.2.1 Interchangeability of landing mat system. Assemble nine full panels and one half panel in three rows as shown in Figure 1 with locking bars. Three full panels and one half panel shall be randomly selected from each 750 panels and assembled as follows: Remove panels 2, 4, 6, and 9 and install the panels selected. Any binding of connectors, misalinement of mat or connector, or failure to assemble locking bars and mat, shall constitute failure of the test (see Figure 1).

4.6.2.2 Structural and reliability test of the mat system. When specified (see 6.2), the landing mat system shall be tested as specified in 4.6.2.2.1 through 4.6.2.2.5. Failure of more than six of the 60 mats scheduled for traffic coverage (see Figure 2) to survive all coverages shall constitute failure of the test. An individual mat failure is defined as any evidence of separation of mats, fractures or deformation resulting in tire safety hazard or cracks exceeding 6 inches in length.

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4.6.2.2.1 Preparation and testing of the subgrade. Construct a section of subgrade not less than 20 feet by 48 feet by 2 feet using a heavy clay soil. Place and compact the soil in 6-inch lifts. The CBR shall be determined in accordance with MIL-STD-161, and the CBR of each lift placed and of the subgrade after construction shall be not less than 3.5 and not more than 4.0. The CBR of the finished subgrade shall be taken at the surface and at depths of 6 and 12 inches at two locations. The two locations shall be 7 feet from and on opposite sides of the longitudinal centerline and one-third of the total distance from each end.

4.6.2.2.2 Installation of mats. Install Type I and II mats on the subgrade as shown on Figure 2. Type III mats will be used as replacement mats for the Type I mats. Additional mats for approach arons may be installed at both ends of the test section (see Figure 2) to form a transition zone. The ends of the mats along the longitudinal edges of the surfaced area shall be anchored with the following anchoring system:

(1)	Anchor Attachment	-C13220E5639
(2)	Edge Anchor Assembly	-C13216E9236
(3)	Bar, Locking	-C13220E5633

4.6.2.2.3 Traffic distribution (coverages). Subject the installed mat to 1000 coverages of an aircraft tire inflated to a pressure of not less than 250 pounds per square inch (psi). The load applied to the tire shall be not less than 50,000 pounds, and the contact area shall be not more than 200 square inches. Apply the load at not more than 4 miles per hour (mph).

4.6.2.2.3.1 Coverage application. Coverages shall be applied in the 11- by 48-foot center of the 36- by 48-foot test section (see Figure 2). Coverage shall be started at one side of the test lane and then back in the same path to complete two coverages. The tire shall then be shifted laterally the width of the tire print, and the procedure repeated a sufficient number of times to cover the width of the test lane. The interior 108 inches of the test lane shall then receive six additional coverages and the innermost 60 inches another two coverages, making a total of 10 coverages for the center 60 inches of the test lane. This cycle shall constitute 10 coverages. Subsequent coverages shall be applied by repeating this cycle as necessary to obtain the required number of coverages as determined by 4.6.2.2.4.

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4.6.2.2.4 Equivalent coverages. The required equivalent coverages shall be calculated prior to starting the test, and at the completion of an equivalent of 500 coverages. The required coverages is calculated prior to the test using the CBR determined by 4.6.2.2.1 in the following equation:

$$\text{Log}_{10} \text{ Equiv. Coverages} = \frac{140.48}{\sqrt{\frac{6173}{\text{CBR}} - 64}} - 0.65$$

At the completion of an equivalent of 500 coverages, the remaining required coverages shall be calculated by using an average of all 12 CBR values taken as specified in 4.6.2.2.1 and 4.6.2.2.4 in the above equation. The value obtained must be divided by 2 to obtain the remaining coverages required. If the subgrade must be reconstructed, the coverages remaining shall be determined by using the CBR values obtained from the reconstructed subgrade to obtain coverages required to complete the test. The value obtained from the above equation shall be divided by 2.

4.6.2.2.5 CBR check during coverage application. At the conclusion of an equivalent of 500 coverages, the CBR shall be determined as specified in 4.6.2.2.1 at two locations within the test lane. Failure of the CBR to conform to the range of values specified in 4.6.2.2.1 shall be reason to stop the test and reconstruct the subgrade to the specified CBR. After failure of a mat during coverage application, the CBR of the subgrade directly beneath the failed mat shall be determined at the surface, and at depths of 6 and 12 inches. Failure of the CBR to conform to the range of the CBR values specified in 4.6.2.2.1 shall be reason to stop the test and reconstruct the subgrade to the specified CBR.

4.7 Inspection comparison. The Government may select landing mats and ancillary items at any time during the contract production period and subject these landing mats and ancillary items to the examination specified in 4.6.1 and the test specified in 4.6.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on landing mats and ancillary items selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production landing mats and ancillary items. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison. Failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

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4.7.1 Inspection failure. Failure of any inspection comparison landing mats or ancillary items to meet any requirement specified herein during and as a result of the examination in 4.7 shall be cause for rejection of the inspection comparison and shall be cause for refusal by the Government to continue acceptance of production landing mats or ancillary items until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on landing mats or ancillary items previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all landing mats and ancillary items accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.8 Inspection of preparation for delivery.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. Samples selected in accordance with 4.8.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

116.	Locking bars not furnished for each mat in bundle.
117.	Mat bundle not constructed as specified on Figure 3.
118.	Materials or containers not as specified for packing. Each incorrect material, deviation from figure requirements, or container shall constitute one defect.
119.	Starting connectors not packed and strapped as specified.
120.	Access adapters not packed and strapped as specified.
121.	Turn adapters, Male I-Lock, not packed and strapped as specified.
122.	Turn adapter, Female I-Lock, not packed and strapped as specified.

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123. Dunnage not meeting requirements of Figure 8.
124. Edge anchor I-Lock attachment not packed and strapped as specified.
125. Edge anchors not packed and strapped as specified.
126. Marking missing, illegible, incorrect or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Preproduction pack. The supplier shall furnish a preproduction pack for examination within the time frame specified (see 6.2) to prove, prior to starting production packing, that the packing and marking comply with the requirements of this specification. Examination shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.7).

5.2 Preservation and packaging. No preservation and packaging shall be required.

5.3 Packing. (See 6.8.)

5.3.1 Truss web mats. The truss web mats with locking bars (hereinafter referred to as "mats") shall be bundled and crated 18 mats per pack in accordance with Figures 3 and 4. A suitable jig shall be used by the supplier to aid in preparing the bundle shown on Figure 3, to assure exact and uniform positioning of the mats. After securing with two straps, the bundle can be moved and positioned at the crate assembly area. Thirty-six half mats secured in pairs with the locking bar as well as 18 repair mats shall be assembled as described above.

5.3.2 Starting connectors. Starting connectors (see Figure 5A) shall be packed flat, 4 per layer with every other connector inverted, 3 layers per box, in a snug-fitting box approximating 9 feet 1/16 inch in length, 11-15/16 inches in width and 4-1/2 inches in depth (inside dimensions) (see 6.9 and Figure 7). Box shall be strapped around the circumference (ends and sides), and then strapped girthwise at each end plus 3 straps between the end straps, equally spaced (total of 6 straps). Strapping shall conform to requirements of QQ-S-781, Type I or IV, Class B, 3/4 X 0.025 minimum.

5.3.3 Access adapters. Access adapters (see Figure 5B) shall be locked together in groups of 7 and packed 7 per layer, 4 layers per box, in a snug-fitting box approximating 9 feet 1/8 inch in length, 12-1/2 inches in width and 6-1/8 inches in depth (inside dimensions). Box shall conform to requirements of Figure 7 and be strapped as described in 5.3.2 and Figure 7 (see 6.9).

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5.3.4 Turn adapter, I-lock, male. Seventy male I-locks along with 70 locking bars (see Figures 5C and 5E), shall be packed in a snug-fitting box conforming to the requirements of PPP-B-601, Overseas Type, Style I (with skids). Three girthwise straps shall be used. A box to contain the above quantities of components, arranged in accordance with the configuration shown in Figure 8, will have inside dimensions approximating 24 inches in length, 23-7/8 inches in width, 8-5/8 inches in depth. The skids, as well as the load components, are to be located parallel with the 23-7/8 inch dimension. Box shall be constructed with the filler panel parallel with the skids. Use filler cleats in the panels perpendicular to the skids. Three straps located parallel with the skids and conforming to QQ-S-781, Type I or IV, Class B, 3/4 X 0.025 inches shall be used.

5.3.5 Turn adapter, I-lock, female. One hundred and twelve female I-locks along with 111 locking bars (see Figures 5D and 5E) shall be packed and strapped in a box as required in 5.3.4. A box to contain the above quantities of components, arranged in accordance with the configuration shown in Figure 9, will have inside dimensions approximating 24-7/16 inches in length, 23-7/8 inches in width, and 10-9/16 inches in depth (see 6.9).

5.3.6 Locking bars. Locking bars (see Figure 5E) are not packed separately, but with the item with which they are to be used (see 5.3.1, 5.3.4, and 5.3.5).

5.3.7 Edge anchor I-lock attachment. Two hundred and eighty edge anchor I-lock attachments shall be packed in a box as required in 5.3.4. A box to contain 288 of this component, arranged in accordance with the configuration shown in Figure 10, will have inside dimensions approximating 30-1/16 inches in length, 26-3/4 inches in width, and 8-1/2 inches in depth (see 6.9.1). Skids are to run parallel with the width dimension.

5.3.8 Edge anchor. One hundred and forth-four edge anchors are to be packed in the arrangement and box construction shown in Figure 11. The box shall conform to PPP-B-601, Overseas Type, Style I. The arrangement of the anchors calls for a row of 8 on top of a row of 10. The following row of 10 shall be so arranged that the eye end of the anchor in the bottom row will be directly under the helix end of the anchor in the third tier, etc. It has been found that the final 16th tier of 8 loaded into the box may fit best if all anchors are laid in one direction. When fully loaded, the box will contain 8 tiers of 10 anchors each, and 8 tiers of 8 anchors each (see 6.9).

5.4 Marking. Marking shall be in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The landing mats and ancillary items, when assembled, comprise an expedient surfacing system which is capable of supporting the operation of the F-111 and C130 type aircraft. The system is capable of sustaining the impact and braking forces of aircraft which have single wheel loads up to 50,000 pounds.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the preproduction model, and types of the mat and ancillary items required (see 3.2).
(c) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some or all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the supplier (see 3.2).
(d) When initial production inspection is required and the number of each type mat and ancillary item to be furnished (see 3.3).
(e) When the structural test is required and whether or not the Government will conduct the structural test (see 4.6.2.2), the contracting officer should specify the time frame for conducting the test.
(f) Time frame required for submission of the preproduction pack (see 5.1).

6.3 Preproduction model. Any changes or deviations of production landing mats or ancillary items from the approved preproduction model during production will be subject to approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish landing mats or ancillary items conforming to this specification.

6.4 Equivalent sorties. The 1000 coverages applied to the mat during the coverage test is equivalent to 5,260 sorties. A sortie is one take off and one landing.

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6.5 Supplementary quality assurance provisions (SQAP). The contracting officer should require the supplier to maintain records of all SQAP inspections. A suggested paragraph is as follows:

> "The supplier shall maintain complete records of all examinations and tests performed to verify the requirements of classified SQAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.6 Definitions.

6.6.1 Supplementary quality assurance provisions (SQAP). A SQAP is a contractual requirement that supplements Section 4 of the specification. SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.7 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligation to pack and mark the landing mats and accessories in accordance with this specification (see 5.1).

6.8 Level of packing (see 5.3). Only a single method of packing is suitable and adequate for this commodity. For purposes of the packing level, marking Level A shall be used. Level C packing is not applicable to this item as the shipping container will also be used in the field as a storage container. As such it is susceptible to exposure to all environmental conditions in addition to rough handling. A lesser level of pack would require a repack prior to troop issue.

6.9 Weight and cube of component packs.

Component Pack & Quantity	Weight (Lbs) Net	Gross	Cube
a. Truss web mat - 18*	2088	2180	59.7
b. Starting connectors - 12	384	457	7.6
c. Access adapters - 28	630	707	9.2
d. Turn adapters, I-lock, male - 70*	315	356	4.9
e. Turn adapters, I-lock, female - 112*	414	456	5.8
f. Edge anchor, I-lock - 288	474	525	6.5
g. Edge anchor - 144	576	655	13.2

*Packed with locking bars.

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6.9.1 Box dimensions. Box dimensions given in Section 5 were based on actual packing of components of previous procurement. As such, succeeding procurements, while still conforming to specification requirements, could have slightly different dimensions when packed as specified in Section 5. Therefore, a preproduction pack should be made before boxes are obtained for the entire procurement quantity.

Custodians:

Army - ME
Navy - YD
Air Force - 84

User activities:

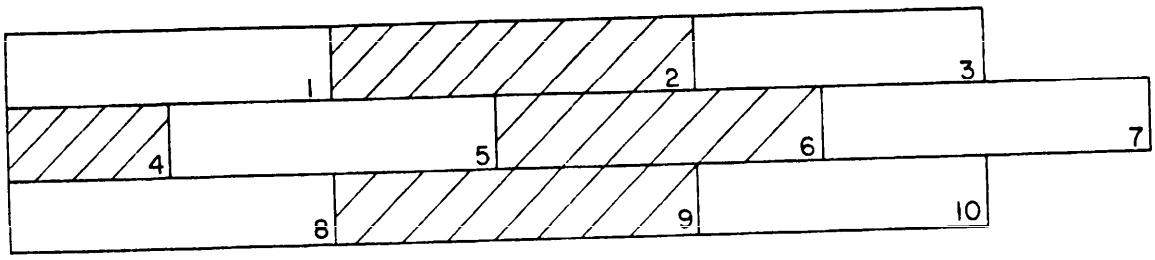
Army - CE
Navy - AS, MC

Preparing activity:

Army - ME

Project No. 5680-0067

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(1) REMOVE PANELS 2, 4 (HALF PANEL), 6, AND 9.

(2) INSTALL SELECTED PANELS IN THESE VACATED POSITIONS AND CHECK FOR FIT.

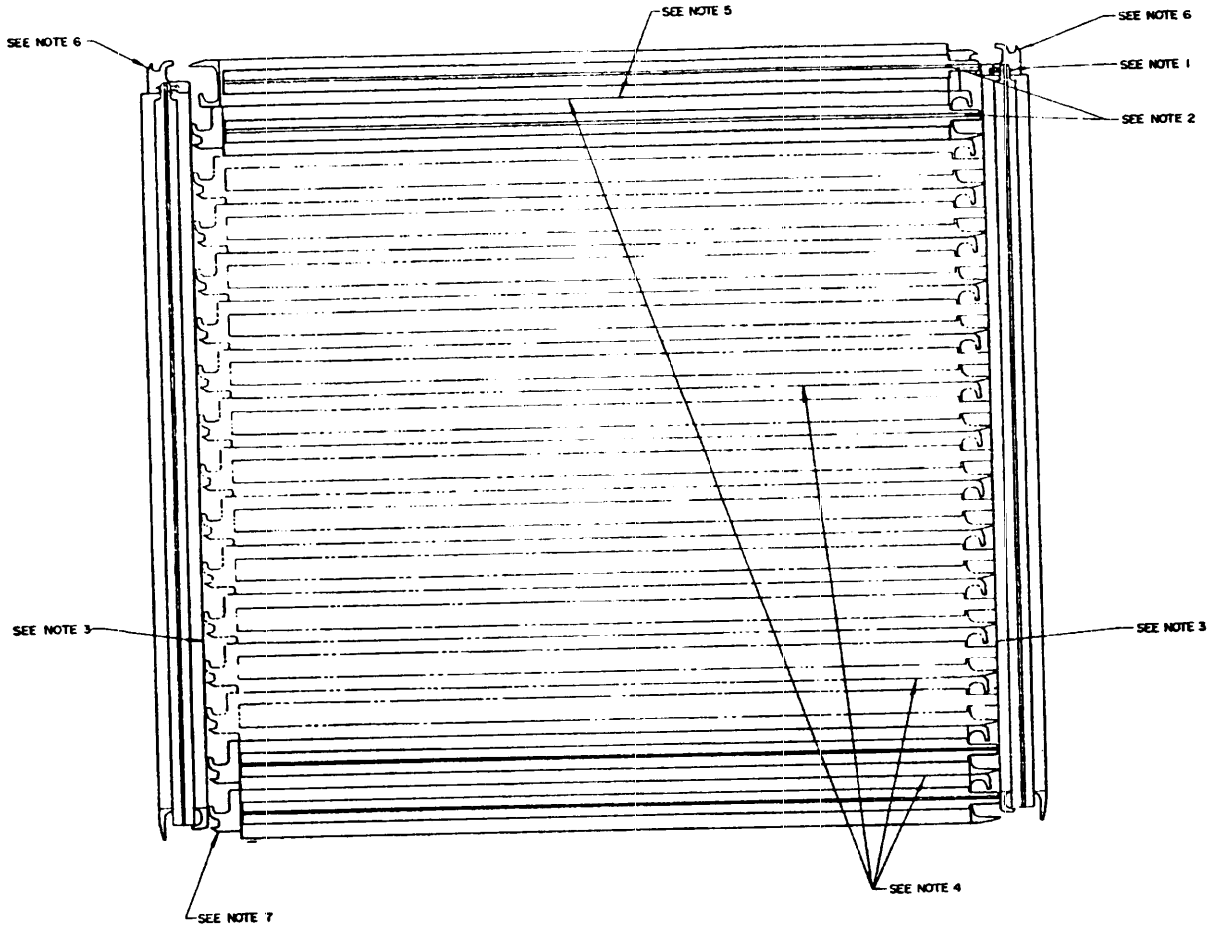
STATIONARY PANELS

PANELS TO BE REMOVED

FIGURE 1. INTERCHANGEABILITY CHECK

X-2942

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NOTES:

1. THE VERTICAL LOCKING BARS ARE TO BE HELD IN POSITION PRIOR TO CRATING THE BUNDLE WITH A 1-IN., NO 12 WOOD SCREW INSERTED THRU THE HOLE IN THE LOCKING BAR
2. LOCKING BARS ATTACHED TO ALL MATS
3. TOP SIDE OF MAT TO INSIDE OF BUNDLE
4. TOP SIDE OF MAT UP
5. TOP SIDE OF LAST MAT DOWN
6. MALE CONNECTOR AT TOP
7. MALE CONNECTOR ALL ON ONE SIDE EXCEPT FOR TOP PANEL.
8. 18 FULL MATS, 36 HALF MATS OR 18 REPAIR MATS POSITIONED IN THIS CONFIGURATION PRIOR TO STRAPPING WITH TWO GIRTHWISE STRAPS, LOCATED APPROXIMATELY 1/4 DISTANCE IN FROM EACH END. STRAPPING TO CONFORM TO QQ-S-781, TYPE I OR IV, CLASS B, 3/4" x 0.025. THIS FIGURE ALSO APPLIES TO 18 REPAIR MATS OR 36 HALF MATS.

FIGURE 3.

PREPARATION OF MAT BUNDLE

X-2963

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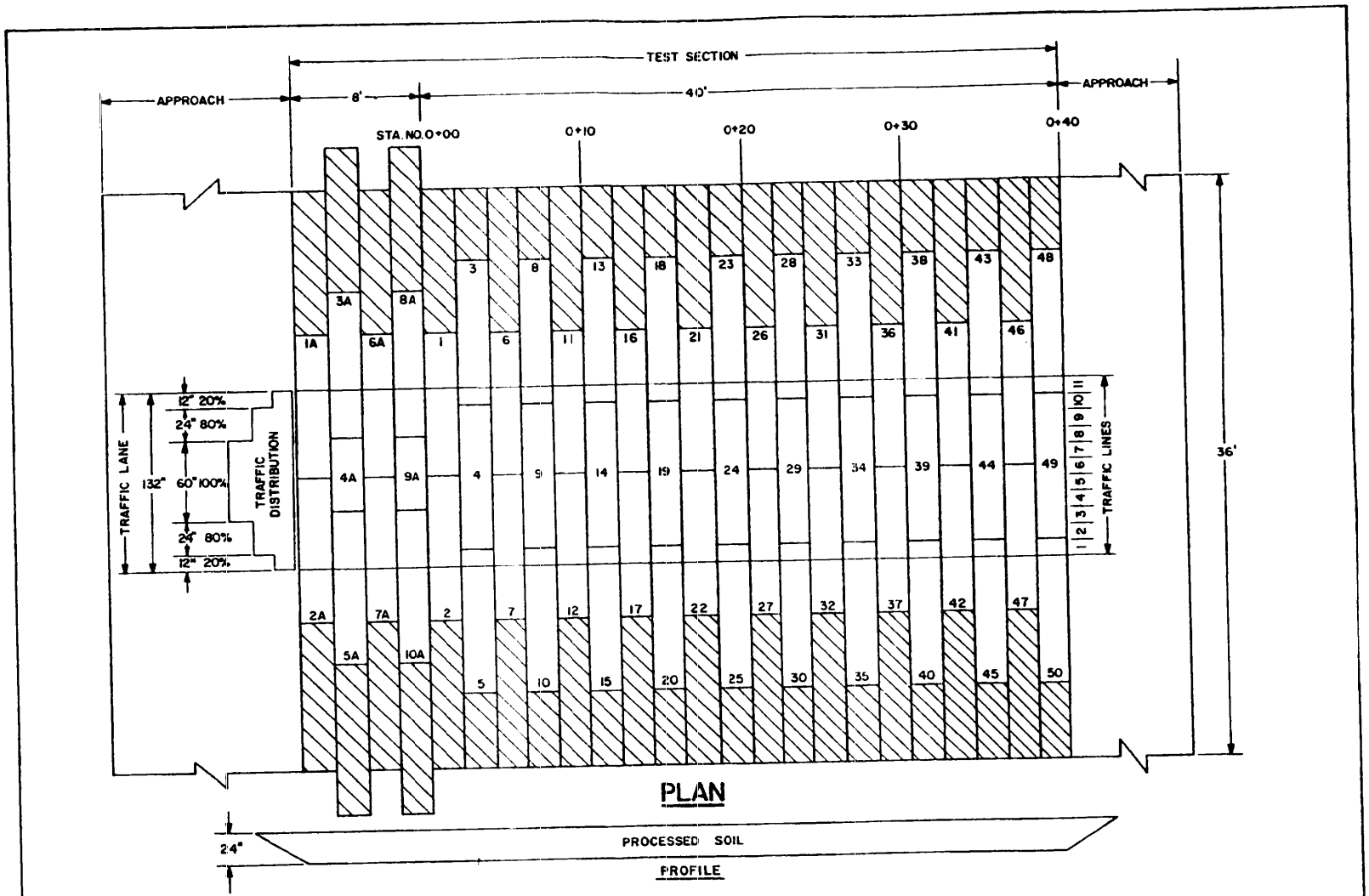
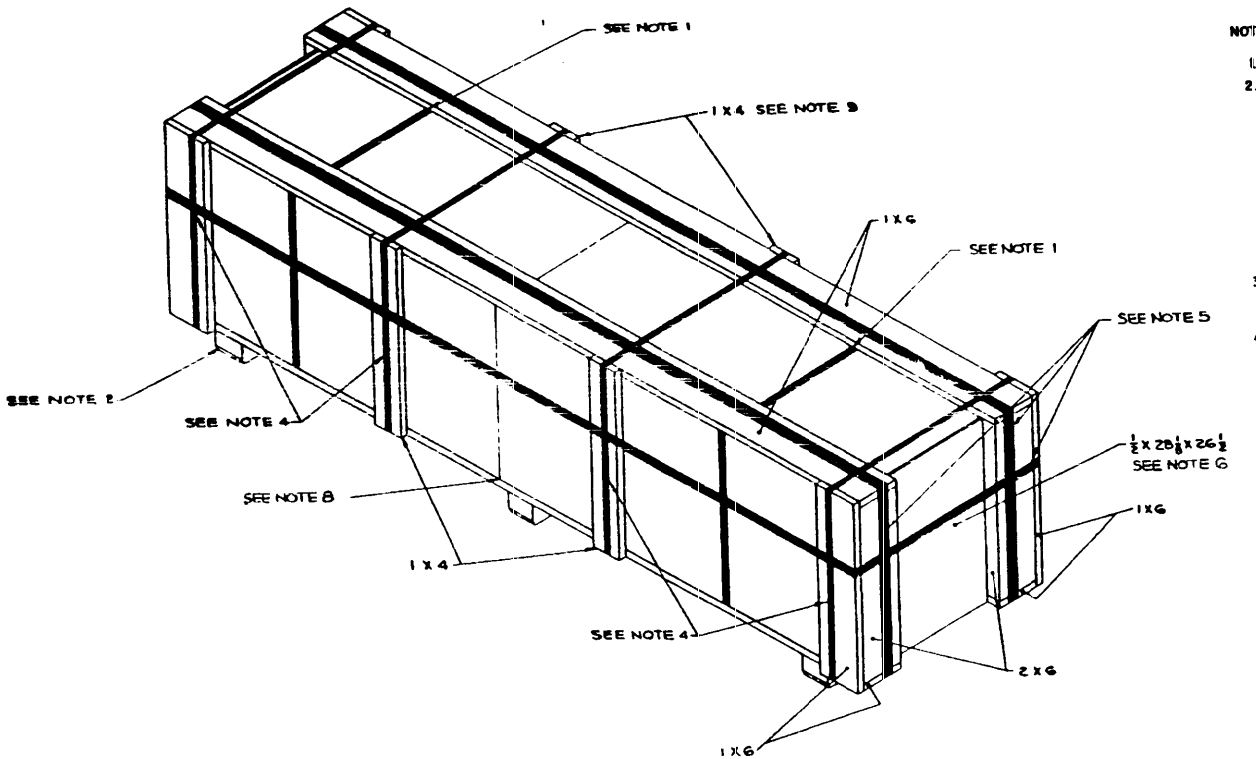


FIGURE 2. STRUCTURAL TEST LAYOUT AND STANDARD TRAFFIC DISTRIBUTION

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NOTES:

1. BUNDLE STRAPPING BEFORE CRATING.
2. NOM 3X4 SKIDS CHAMFERED ON CRATE SIDES. SKIDS SECURED TO 1X6 MEMBERS WITH NAILS CONFORMING TO FF-N-105, TYPE II, STYLE 18, 2 1/2" LG (MIN) NAILED THRU CRATE MEMBER INTO SKID, 3 NAILS PER CRATE MEMBER, PER SKID. SKIDS TO BE NOTCHED FOR TWO LONGITUDINAL STRAPS, 1 1/2-INCHES X 1/4-INCH, STARTING 2 INCHES FROM END. SKIDS TO BE MOUNTED TO BOTTOM 6 INCHES IN FROM ENDS AND IN CENTER.
3. ALL NAILS, EXCEPT FOR SECURING SKIDS, TO CONFORM TO FF-N-105, TYPE II, STYLE 7, 8 OR 9, CEMENT CORED, 7d MINIMUM.
4. STRAPPING APPLIED OVER BATTENS CONFORMING TO QQ-S-781, TYPE I OR II, CLASS B, 3/4 X 0.025.
5. LENGTHWISE STRAPPING APPLIED AS SHOWN TO CONFORM TO QQ-S-781, TYPE I OR II, CLASS B, 1 1/4 X 0.035.
6. PLYWOOD ENDS TO CONFORM TO NN-P-530, GROUP A (PS-51, TYPE I, GRADE 3-4) OR GROUP B (PS-1), STANDARD SHEATHING, WITH EXTERIOR GLUE, 1/2 INCH THK. SIZE OF PLYWOOD WILL BE DEPENDANT ON FINISHED MAT TO INCLUDE THICKNESS OF WELDS.
7. WOOD MEMBERS OF CRATE TO CONFORM TO MIL-STD-731, WOOD, GROUP II OR III, GROUP 2.
8. VIEW WHEN 36 HALF MATS ARE CRATED.
9. LENGTH OF 1X4 BATTENS WILL BE LESS THAN THAT OF THE 1X6 MEMBERS. MAT BUNDLE WILL SAG IN CENTER DUE TO END WELDS.

FIGURE 4.
CRATING DETAILS FOR MAT BUNDLE
APPLICABLE TO WHOLE MATS, HALF MATS AND REPAIR MATS

X-2964

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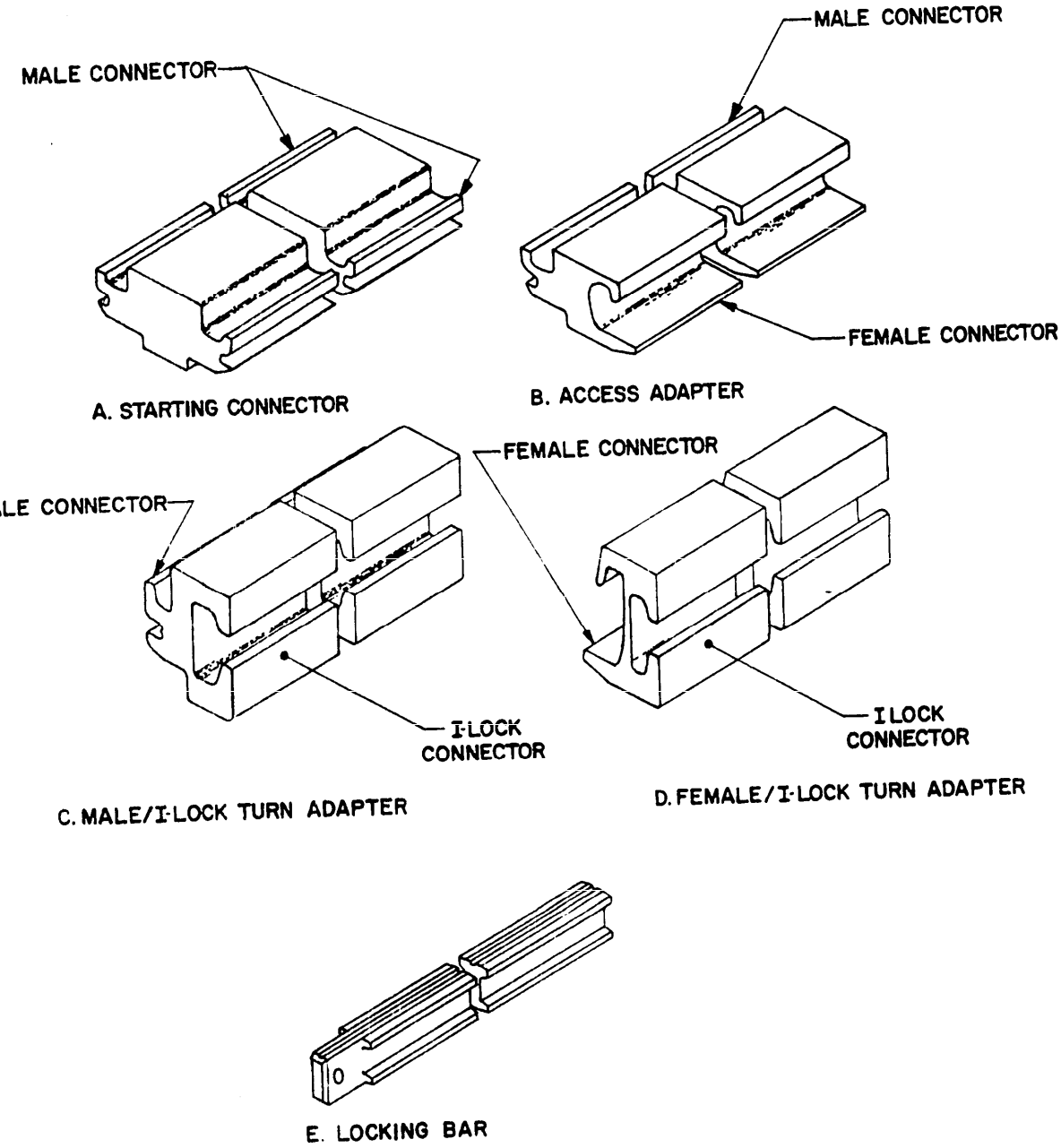
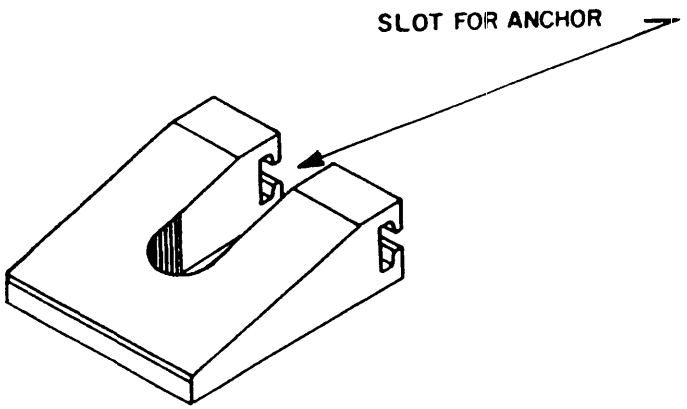


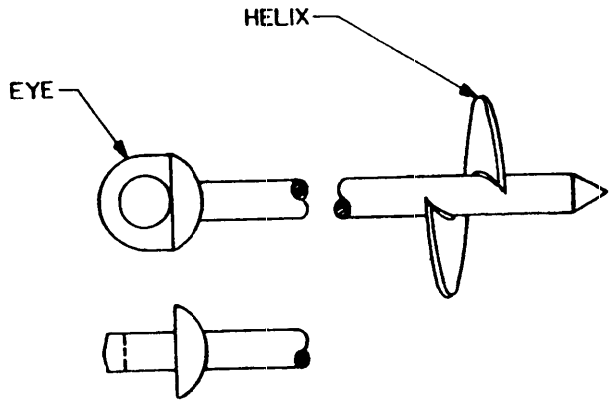
FIGURE 5.
VARIOUS ANCILLARY ITEMS

X-2965

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A. I- LOCK ATTACHMENT



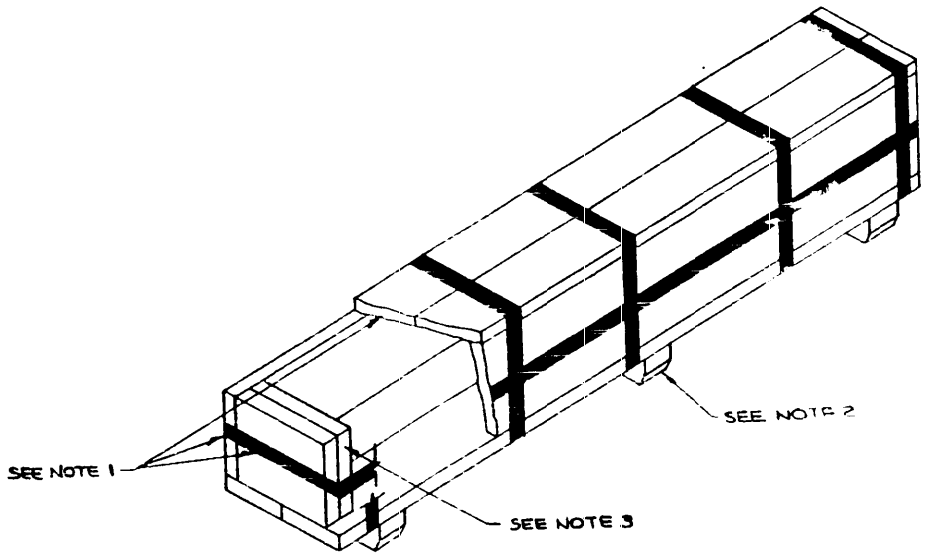
B. EDGE ANCHOR

FIGURE 6.

EDGE ANCHOR AND ANCHOR I-LOCK ATTACHMENT

X-2966

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NOTES:

1. SIDES, ENDS, TOP AND BOTTOM SHALL BE NOMINAL 1-INCH LUMBER CONFORMING TO REQUIREMENTS OF MIL-STD-731, WOOD, GROUP II OR III, CLASS 2.
2. SKIDS SHALL BE NOMINAL 3 X 4, CHAMFERED ON BOX SIDES. SKIDS SHALL CONFORM TO REQUIREMENTS OF NOTE 1 ABOVE, EXCEPT THAT CLASS 3 MATERIAL MAY BE USED. SEE FIG. 4 FOR SKID NAILING REQUIREMENTS.
3. INNER PORTION OF THE DOUBLE END BOX SHALL BE 3/4 INCH THICK PLYWOOD CONFORMING TO NOTE 6, FIG. 4.
4. NAILING AT THE ENDS SHALL BE INTO THE OUTER WOOD PORTION OF THE DOUBLE END. NAILS SHALL COMPLY WITH REQUIREMENTS OF NOTE 3, FIG. 4.

FIGURE 7.

CONSTRUCTION AND STRAPPING DETAILS,
STARTING CONNECTORS AND ACCESS ADAPTERS X-2967

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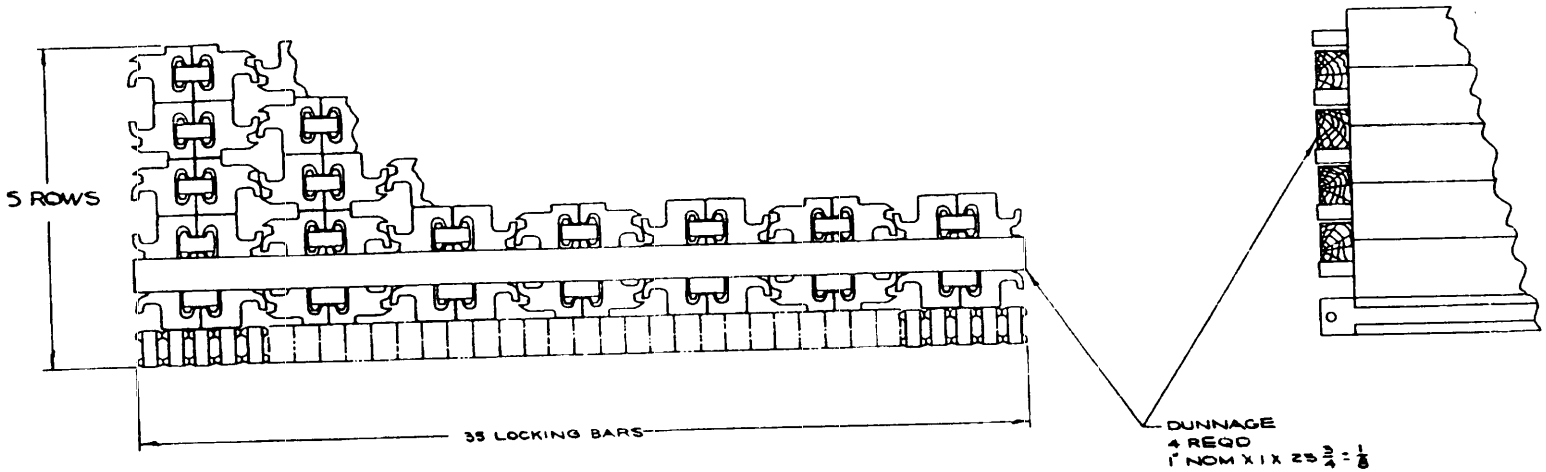
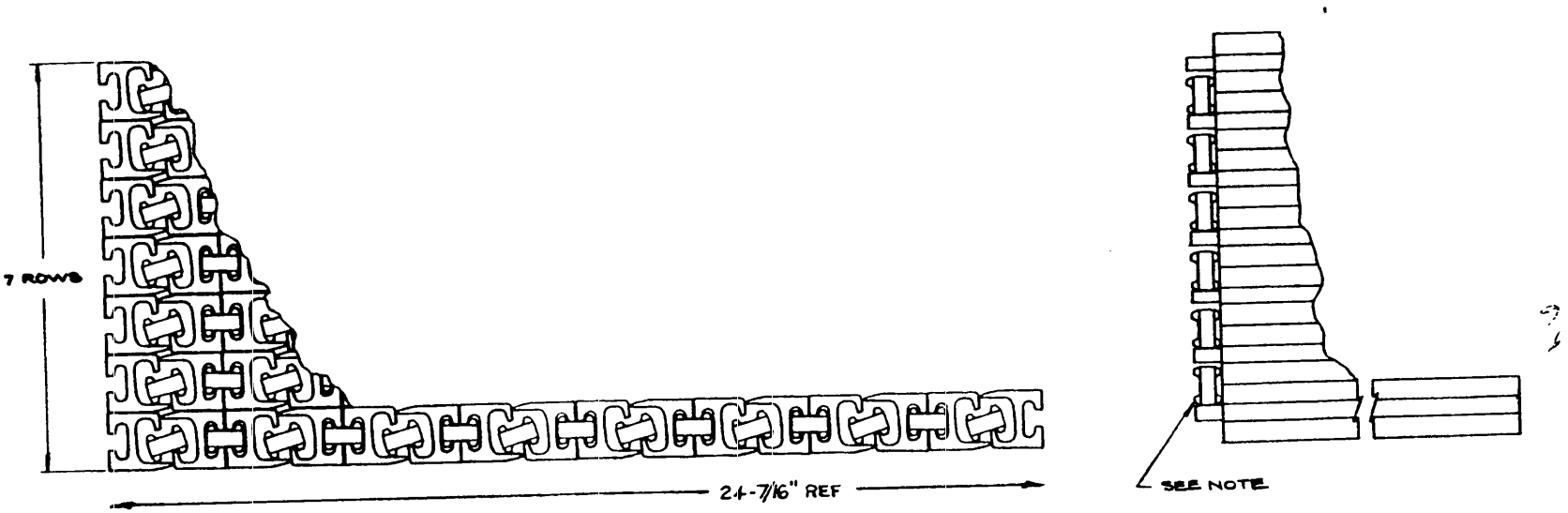


FIGURE 9.
PACKING CONFIGURATION FOR TURN ADAPTER, I-LOCK, MALE, 70 EACH ALONG WITH 70 LOCKING BARS

X-2969

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NOTE: LOCKING BAR SHOWN ON END VIEW REMOVED FROM FRONT VIEW FOR CLARITY

FIGURE 8.
PACKING CONFIGURATION FOR TURN ADAPTER I-LOCK, FEMALE, 112 EACH ALONG WITH 111 LOCKING BARS.

X-2968

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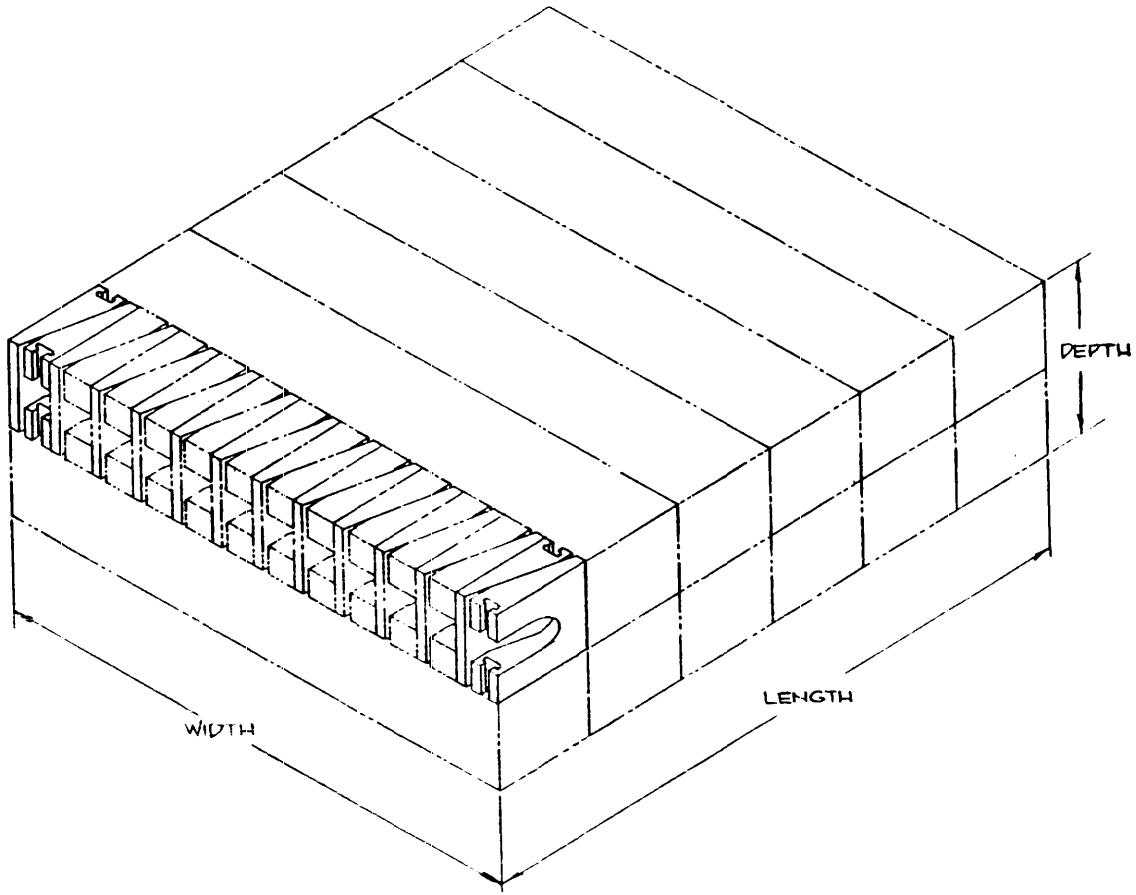
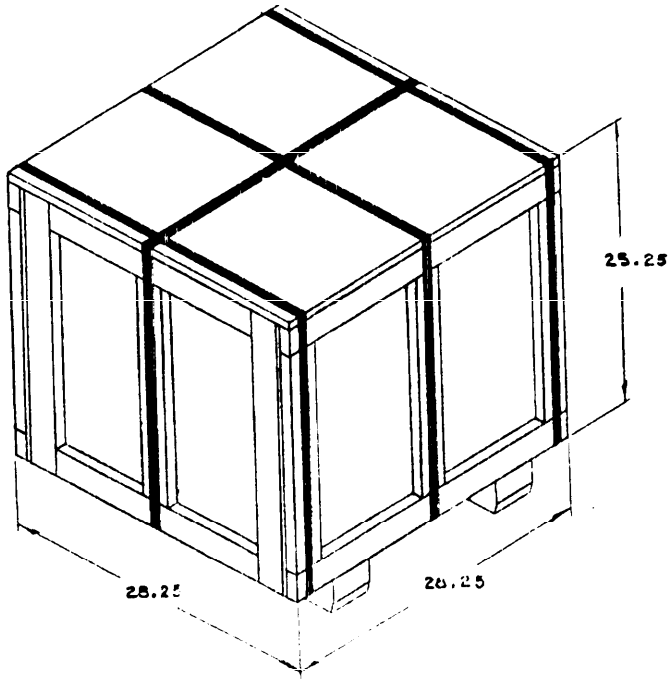


FIGURE 10.

PACKING CONFIGURATION FOR
EDGE ANCHOR I-LOCK ATTACHMENTS, 288 EA.

K-2970

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NOTES:
1. NOTCH SKIDS 1"X 1/4" IN CENTER.
2. STRAPPING TO CONFORM TO QQ-S-781, TYPE I OR II, CLASS B, 3/4 X .025.
3. SKIDS PARALLEL WITH SHORT PANEL.

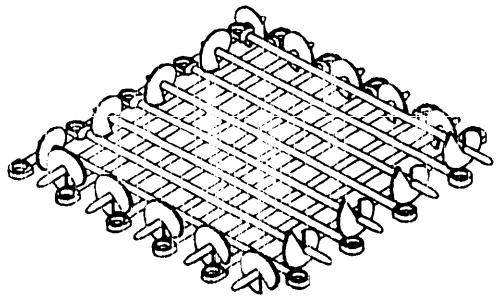


FIG. 11.

BOX DESIGN AND LOAD
CONFIGURATION FOR EDGE ANCHORS

X-2971

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